
Reinforcement Learning from Heterogeneous Demonstrations

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Abstract

Reinforcement Learning, although has plenty of advantages and applications, is time-consuming and sample inefficient in many problems. Learning from Demonstrations, which alone is limited to only reach what it observes, is used together with Reinforcement Learning to improve learning in many applications, including but not limited to robot learning and self driving cars. However, in most of the literature, it is assumed that the provided demonstrations are optimal and if not optimal, are sub optimal, or at least, are performing what our agent wants to do, with different qualities. This is time consuming and costly to obtain in many tasks and requires human effort. In this work, we are going to use a heterogeneous set of demonstrations that are not directly related to our agent's goal, as behavioural policies, in the mentioned applications. In order to do so, we add a new term to the policy loss off Soft Actor Critique method, using Free Energy framework.

1 Introduction

Machine Learning methods are from the most used methods in fields such as robot learning, self driving cars, medication, and etc. Using its methods, human kind was able to explore the deep sea [7] and other planets such as mars [28], assign tasks that are risky or dangerous for humans to autonomous agents [4], save forests from fire [35], clean the oceans [2], and in general improve humans life [29]. Although machine learning methods have improved our life greatly, they are still lacking compared to humans in performing everyday tasks in a real environment and need further improvements.

One of the most important method among machine learning methods is reinforcement learning, which is inspired from human and animal learning [11]. The goal of the reinforcement learning agent is to maximize its expected reward, which is appropriate for many applications including robot learning [21], game playing [40], and self driving cars [37], among many other. Allowing agents to learn through interacting with the environment [26], reinforcement Learning is considerably successful in addressing decision-making problems [26]. However, it still has some disadvantages, including being time-consuming for complex problems, unable to reach the optimal result, sample inefficiency, and dependency to reward shaping.

Inspired from humans and animals, learning from demonstration is used to address the mentioned problems and reduce the search space of reinforcement learning methods [5]. Learning from demonstration is well studied in the literature both individually and merged with reinforcement learning, but in most of the literature, the demonstrations are assumed to be optimal or at least sub-optimal, and directly relevant to what our agent wants to achieve. Gathering this sort of demonstrations is costly and time-consuming and in some cases not even possible. Thus, in this work we want to address this problem, and use diverse demonstrations to improve the performance of reinforcement learning.

We assume that each demonstrator's policy is a point in the policy space that can guide us to improve our learning, similar to Figure 1, and we will choose a behaviour policy to bias our decisions towards it. We use the idea of Free Energy frame work [32], and evaluate different methods of finding a behavioural policy that decreases the cost of policy search. We show that assuming the demonstrator

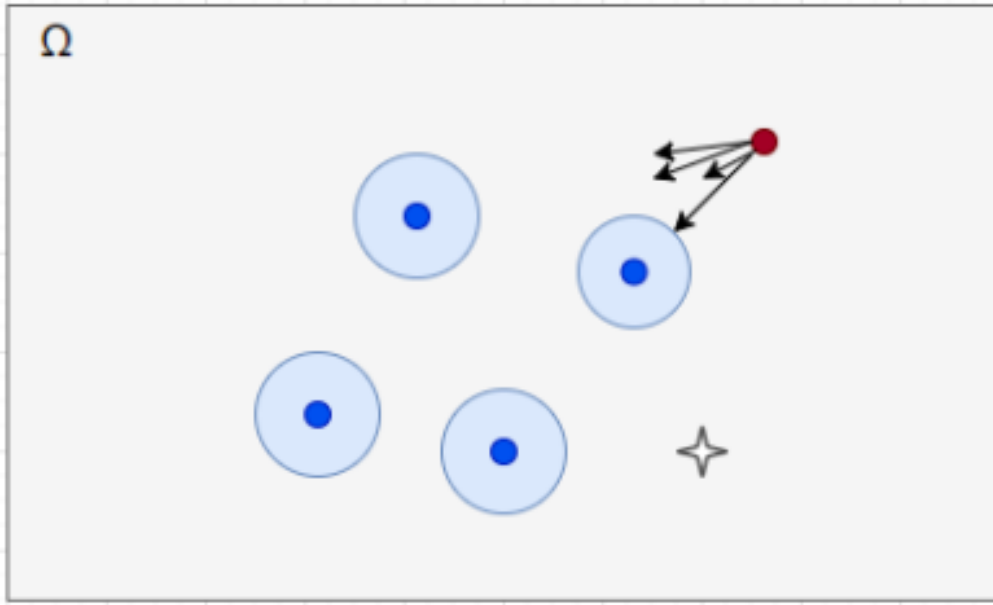


Figure 1: Our policy, the red dot, the demonstrator’s policy, the blue areas, and the optimal policy, the star, in the policy space.

with maximum evaluated return, evaluated using off policy evaluation methods, and update the behavioural policy, reduces the search cost considerably.

2 Related work

Demonstrations are used for more than three decades for learning problems, especially in fields such as robotics. At first, remotely controlling robots was used to gather demonstrations [27, 14, 25, 38]. This remote controlling was hard and needed expertise in robot remote controlling, thus other methods were proposed such as moving robot’s body manually or using computer vision [24, 23, 20].

At first demonstrations were used alone to only produce the exact behaviour of the demonstrators, but this in fact had many disadvantages. One of the disadvantages comes from the fact that demonstrations can not cover the whole state space. In addition to that the quality of demonstrations influenced the performance of the agent greatly. The idea to solve these problems was to use other methods such as Reinforcement Learning alongside learning from demonstrations [3].

[39] used a two phased method to merge reinforcement learning with learning from demonstrations. At the first phase a teacher selects the appropriate actions and the reinforcement learning module trained on the selected actions. The second phase selects the actions according to reinforcement learning module and continues to learn. In addition, [41] records the policies using remotely controlling the agent or applying a sub-optimal policy and extracts some rules from the demonstrations. Then the agent decides online whether to use these rules or use its internal reinforcement learning.

Reinforcement learning could also use learning from demonstrations to improve its performance. One of the most important disadvantages of reinforcement learning methods are being sample inefficient and time-consuming. Similar to [41], in [34] reinforcement learning is used together with learning from demonstrations, but with the goal of decreasing the learning time of reinforcement learning. The error of imitation learning is added to the error of reinforcement learning in [30] and when the reinforcement reached the performance of demonstrations this error will no longer be used.

[36] also used states of demonstrations as the starting states and [6] biases its decision to demonstrations using shaping reward function. In [1] appropriate representations are learned from demonstrations and reinforcement learning is used to continue learning with the learned demonstrations. Other methods such as pre-processing using demonstrations [13] and filling memory with demonstrations [18, 43] are used to improve the learning pace of reinforcement learning algorithms.

Another disadvantage of reinforcement learning is being unable to find optimal policy when the

state space is large. In [10] the states that look irrelevant to the agents goal are extracted from demonstrations to decrease the search space. In addition, sparse reward environment are difficult for reinforcement learning agents to solve, which is addressed in [19] using encouraging the agent to explore states similar to observations. [42] also adds causality to reinforcement learning from demonstrations to solve sparse reward problems.

Another method to solve disadvantages of reinforcement learning is through using hierarchical reinforcement learning, which has its own disadvantages such as needing for the sub goals and options to be defined manually. [9] used demonstrations to find sub goals and [45, 22, 15] use them to find options.

In recent methods, there are few ones who address improving reinforcement learning from non optimal demonstration. [33] uses a method named self imitation learning [31], which imitates from its past experiences, to imitate demonstrations and shows that it can work well for sub optimal demonstrations. [46] also uses demonstrations but improves it using reinforcement learning to address issues of sub optimal demonstrations. [12] assumes demonstrations to have different point of view and tries to use them to improve its learning.

No work in the literature aimed to use heterogeneous demonstrations. However, algorithms proposed in [8], [44] and [16] might also work for heterogeneous demonstrations, which we will analyse in this work. [8] assumes that all demonstrations are from the same task but with different quality. The method proposed is to influence demonstrations by their correspondence weights per each state action pair. [44] creates an ensemble of demonstrations and adds pairs that have high weight to the ensemble. Finally, [16] breaks demonstrations as sub demonstrations and uses them as new demonstrations. Although it is not mentioned or evaluated in the relative paper, the mentioned methods might work for heterogeneous demonstrations and we need to compare our method with them.

3 Problem Formulation and Idea

In this section, we are going to mathematically formulate the problem that we intend to address in our further studies. We consider a Markov decision process (MDP), defined by $\langle S, A, P, R, \gamma, \mu \rangle$, in which $R : S \times A \rightarrow \mathbb{R}$ is the reward function, $P : S \times A \times S \rightarrow [0, 1]$ is the transition function, $\gamma \in (0, 1)$ is the discount factor, μ is initial state distribution, and S and A denote the state space and action space respectively that are defined for discrete and continuous cases as follows:

$$\text{Discrete} : S = \{s_i; i \in (0, N)\}, A = \{a_i; i \in (0, M)\}$$

$$\text{Continuous} : S = \langle s_1, \dots, s_d \rangle, s_i \in \mathbb{R}, A = \langle a_1, \dots, a_q \rangle, a_i \in \mathbb{R}$$

A policy $\pi : S \times A \rightarrow [0, 1]$ is map from a state to action probabilities and a trajectory $\tau = \langle s_0, s_1, a_1, \dots, s_T, a_T \rangle$, where T denotes terminal state, can be generated from π . Reinforcement learning algorithms try to find an optimal policy π^* that maximizes the discounted return, i.e.:

$$\pi^* = \underset{\pi}{\operatorname{argmax}} \mathbb{E} \left[\sum_{t:1 \text{ to } T} \gamma^t R(s_t, a_t) \right].$$

Our agent has initially access to pairs of heterogeneous demonstrations ($Demo = \{demo_i; i \in (0, D)\}$) collected from multiple agents performing multiple tasks. These demonstrations can be considered as the input of our problem or they can be derived from a policy or a data set of videos, $V = \{v_1, \dots, v_k\}$, where v_i is video of an agent performing a task. We first assume that the policy of demonstrators are extracted from the demonstrations, $\Pi = \{\pi_{d1}, \dots, \pi_{dk}\}$, and these demonstrations are from another MDP with isomorphic transition function P , state space and action space, but different reward function. We can further analyse the case that state space and action space is also different.

Our main idea is to minimize the information cost through minimizing the distance of our policy to a behaviour policy, in addition to maximizing the expected utility, similar to Free Energy methods. We do this by adding a new term to the policy loss of Soft Actor Critique, which is an off-policy method and tries to maximize the expected utility.

4 Preliminaries

4.1 Soft Actor Critic

Soft Actor Critic or SAC is an off policy policy gradient method that uses value function approximation to better train the policy [17]. SAC maximizes long-term rewards in addition to action

randomness which the objective function can be formulated as:

$$J_{\pi}(\phi) = \sum E_{(s_t, a_t) \pi_{\phi}} [r(s_t, a_t) + \alpha * H(\pi(.|s_t))]$$

Where H is entropy which is controlled using α . The method has a policy network, that returns mean and variance which the actions are sampled from. The update of the policy network is based on Q value estimations, which is itself another network. The Q value update is also related to value function which is again estimated using another network. The objective function of policy network, Q network and value network are as follows:

$$J_V(\phi) = E_{S_t \sim D} [\frac{1}{2} * (V_{\psi}(s_t) - E_{a_t \sim \pi_{\phi}} [Q_{\theta}(s_t, a_t) - \log \pi_{\phi}(a_t|s_t)])^2]$$

$$J_Q(\theta) = E_{(s_t, a_t) \sim D} [\frac{1}{2} (Q_{\theta}(s_t, a_t) - r(s_t, a_t) - \gamma * E_{s_{t+1} \sim p} [V_{\psi}(s_{t+1})])^2]$$

$$J_{\pi}(\phi) = \nabla_{\phi} E_{S_t \sim D, a_t \sim \pi_{\phi}} [D_{KL}(\pi_{\phi} || \frac{\exp(Q_{\theta}(s_t, .))}{Z_{\theta}})]$$

4.2 Free Energy

Free energy is used as the objective function for bounded-rational decision-makings and is used to model information constrained decision-making. The goal in it is to minimize the information cost while maximizing the expected utility as follows:

$$\pi^*(a|s) = \operatorname{argmin}_{\pi(a|s)} (E_{\pi(a|s)} [\frac{1}{\alpha} \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} - U(a, s)])$$

Where $U(s, a)$ is the utility function, $\pi_{\beta}(a|s)$ is our policy that we are learning, α is the inverse temperature that controls the trade-off between information cost and maximizing utility, and π_{β} is the behaviour policy, or prior knowledge.

5 Proposed Method

The other idea is to use the demonstrators policy using Free Energy Minimization method [32]. We could assume that the policies we obtained from observations are points on the policy space that can guide us to the optimal policy. As we mentioned before, the free energy of a probability distribution P is defined as:

$$F_{\alpha}(\pi(a, s)) = E_{\pi(a|s)} [\frac{1}{\alpha} \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} - U(a, s)]$$

In reinforcement learning $\pi(a, s)$ can be a representative of our policy and $\pi_{\beta}(a|s)$ can be assumed as a behavioural policy. If we had one demonstrator we could have assumed its policy as $\pi_{\beta}(a|s)$, but here we have multiple demonstrators.

We could maximize our expected reward similar to SAC using minimizing $D_{KL}(\pi_{\phi} || \frac{\exp(Q_{\theta}(s_t, .))}{Z_{\theta}})$ and define our information cost as the weighted sum of our policy to the policies of the demonstrators or $\sum \lambda_i * D_{KL}(\pi_{\phi} || \pi_i)$. Thus, we change the policy objective function of SAC as follows:

$$J_{\pi}(\phi) = \nabla_{\phi} E_{S_t \sim D, a_t \sim \pi_{\phi}} [D_{KL}(\pi_{\phi} || \frac{\exp(Q_{\theta}(s_t, .))}{Z_{\theta}})] + \sum \lambda_i * D_{KL}(\pi_{\phi} || \pi_i)$$

However, The problem with above formula is that if the other policies are spread across the policy space, one of the KL distance terms could become infinity and decreasing the weights will not prevent the objective function to become infinity. Thus, we need to define or select a behavioural policy using demonstrated policies. We will show in the result that the best method is to use off-policy evaluation and set the policy with maximum evaluated utility as our behavioural policy. In simple words, it means that to choose the policy that is the closest to the optimal policy as our behaviour policy that we want to get close to. Thus, we will write the policy loss as follows:

$$J_{\pi}(\phi) = \nabla_{\phi} E_{S_t \sim D, a_t \sim \pi_{\phi}} [D_{KL}(\pi_{\phi} || \frac{\exp(Q_{\theta}(s_t, .))}{Z_{\theta}})] + D_{KL}(\pi_{\phi} || \pi_{\beta})$$

Where π_β is the policy of the demonstrator with maximum evaluated utility, using off-policy method. As it is shown in Algorithm 1, after rolling out and saving the experience in the common experience buffer, we use the experience to update the demonstrated policies using off policy SAC, and use off policy evaluation to evaluate them. Then we set the demonstrated policy with maximum evaluated utility as the behaviour policy and use the above formula to update our policy.

Algorithm 1 Proposed Algorithm V1

Require: policy of each demonstrator π_i
 initialize $\alpha_i, \phi, \psi, \hat{\psi}, \theta$, common buffer D
for it in $iterations$ **do**
 Roll out policy π_{i_n} k times
 add experience to D
 for each policy π_i **do**
 sample experience from D
 update π_i using SAC
 end for
 sample experience (state, action, nextstate) from D
 $J_V(\phi) = E_{S_t \sim D} [\frac{1}{2} * (V_\psi(s_t) - E_{a_t \sim \pi_\phi} [Q_\theta(s_t, a_t) - \log \pi_{\phi_{hi}}(a_t | s_t)])^2]$
 $J_Q(\theta) = E_{(s_t, a_t) \sim D} [\frac{1}{2} (Q_\theta(s_t, a_t) - r(s_t, a_t) - \gamma * E_{s_{t+1} \sim p} [V_{\hat{\psi}}(s_{t+1})])^2]$
 for each policy π_i **do**
 $EU_i = \exp(\pi_i(s, a)) / \exp(\pi_\phi(s, a)) * reward$
 end for
 $\pi_\beta = \operatorname{argmax}_{EU_i} \pi_i$
 $J_\pi(\phi) = \nabla_\phi E_{S_t \sim D, a_t \sim \pi_\phi} [D_{KL}(\pi_\phi || \frac{\exp(Q_\theta(s_t, .))}{Z_\theta}) + D_{KL}(\pi_\phi || \pi_\beta)]$
 $\psi = \psi - \lambda_V * \nabla_\psi J_V(\psi)$
 $\theta = \theta - \lambda_Q * \nabla_\theta J_Q(\theta)$
 $\phi = \phi - \lambda_\pi * \nabla_\phi J_\pi(\phi)$
 $\hat{\psi} = \tau * \psi + (1 - \tau) * \hat{\psi}$
end for

6 Result

In order to test our method, we need to design our own environments, and we use PyBullet to design the environments. We will discuss these environments in details.

E1: reach_goal_min_energy The agent is a Franka Panda arm, which aims to reach the goal with minimum usage of energy.

E2: reach_goal_avoid_obstacle The same robotic arm, which aims to reach the goal, while avoiding an obstacle.

E3: push_box_to_goal Again, the same agent aims to push a box to the destination.

E4: pick_and_place_box The Franka Panda arm agent, with the purpose of picking a box and placing it in the destination.

We train three demonstrator policies (**E1**, **E2**, **E3**), and use them in the training of the **E4** environment. We let the agent to train for 300 trials and repeat it 10 times. The result of comparing simple SAC method with our method is visible at Figure 2. We can see that our method improved SAC considerably. Also, it is obvious that the SAC was unable to learn the task, considering its inability to achieve the goal, in contrast to our method that was able to successfully learn the pick and place task in limited trials.

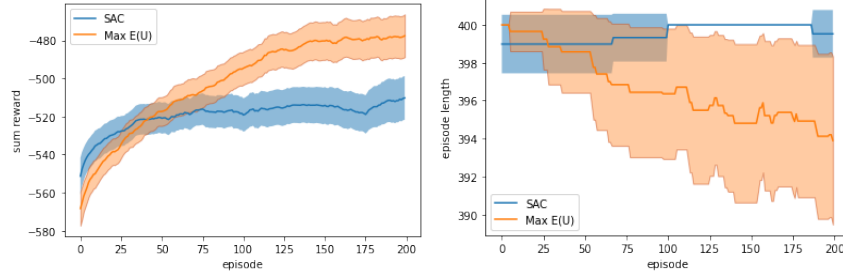


Figure 2: cumulative reward of our method compared to SAC.

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